



REVIEW

Which health benefits of the Mediterranean diet are actually supported?

REFERENCES	TOKENS	VERIFICATION	COMPILED
97 verified	≈19.8k	Crossref	26 Aug 2026

ABSTRACT

Mediterranean dietary programmes have their strongest clinical evidence in cardiovascular prevention. In higher-risk adults, three endpoint trials and modern syntheses indicate fewer major cardiovascular events, although absolute benefits are modest and Mediterranean programmes are not clearly superior to every supported low-fat programme. Randomized prevention, prospective cohort, and biomarker evidence also align for type 2 diabetes, while established disease shows small improvements in glycaemia and body mass. Higher adherence is consistently associated with longer survival, but no mortality trial establishes causality. Cancer and dementia evidence is dominated by cohorts; the largest dedicated cognitive trial was null. Small depression trials are favourable but heterogeneous, and fatty-liver effects resemble low-fat diets unless calorie restriction, activity, or polyphenol enrichment is added. Changes in inflammation, lipids, metabolites, and gut microbial functions cannot substitute for clinical outcomes. Overall, the pattern is a supported cardiometabolic programme whose effectiveness depends on adherence and implementation, not a collection of therapeutic foods or a universal preventive treatment.

01 Introduction

The Mediterranean diet is unusual in nutrition research because several major claims have been tested against clinical endpoints rather than only weight, cholesterol, or self-reported disease. Evidence includes randomized controlled trials of primary and secondary cardiovascular prevention, long prospective follow-up, and mechanistic trials using biomarkers and food provision. Yet the label covers variable interventions, and mortality, cancer, dementia, and mental-health evidence remains largely observational. The central question is not whether the pattern is “healthy” in the abstract, but which outcomes it changes, by how much, and under which comparison conditions. [Karam et al. 2023](#), [Nucci et al. 2026](#), [English et al. 2021](#)

The distinction between a diet and a supported programme is consequential. Endpoint trials supplied foods, dietitian contact, group sessions, and monitoring, while newer studies add energy restriction, physical activity, or behavioural support. [Estruch et al. 2018](#), [Delgado-Lista et al. 2022](#), [Ruiz-Canela et al. 2025](#) Broad dietary claims, by contrast, draw far more often on cohort data and remain vulnerable to healthy-user confounding. [English et al. 2021](#) The throughline is therefore an evidence gradient: causal claims weaken as the review moves from supported cardiometabolic programmes and randomized events to observational associations and surrogate mechanisms.

02 Definition and measurement

The intervention is a family of related eating patterns rather than a prescribed menu. Reviews consistently identify a

plant-rich core—vegetables, fruit, legumes, nuts, minimally refined cereals—combined with olive oil as the main added fat, regular fish, and lower consumption of red or processed meat and sweets. Dairy, eggs, alcohol, portion sizes, and frequencies vary across historical and modern definitions. [Davis et al. 2015](#), [Bach-Faig et al. 2011](#) The Seven Countries Study helped formulate the hypothesis through long-term contrasts among populations, but its ecological and cohort observations were not randomized evidence. [Menotti & Puddu 2015](#)

Most studies convert the pattern into an adherence score. Some scores award points relative to sex-specific or study-specific medians, while others use fixed intake targets; trials often use the 14-item Mediterranean Diet Adherence Screener (MEDAS). [Bach et al. 2006](#), [Schröder et al. 2011](#) These tools make a whole-diet exposure tractable, but they are not interchangeable: a two-point difference can represent different food substitutions in different datasets, and scoring systems developed in Mediterranean populations may classify non-Mediterranean diets differently. [Hutchins-Wiese et al. 2022](#) Direct comparison within the MCC-Spain study found limited agreement among adherence indexes, showing that instrument choice can change who is labelled highly adherent. [Olmedo-Requena et al. 2019](#)

Synthesis quality compounds this problem. Across 24 cardiovascular reviews, complete satisfaction of adapted methodological criteria ranged from 8% to 75%; search details, participant characteristics, or appropriate pooling methods were often missing. [Huedo-Medina et al. 2016](#) That label

does not resolve variation in diet definition, comparator, support, or outcome measurement.

The stable core, variable definitions, and unsupported shortcuts are separated in Figure 1.

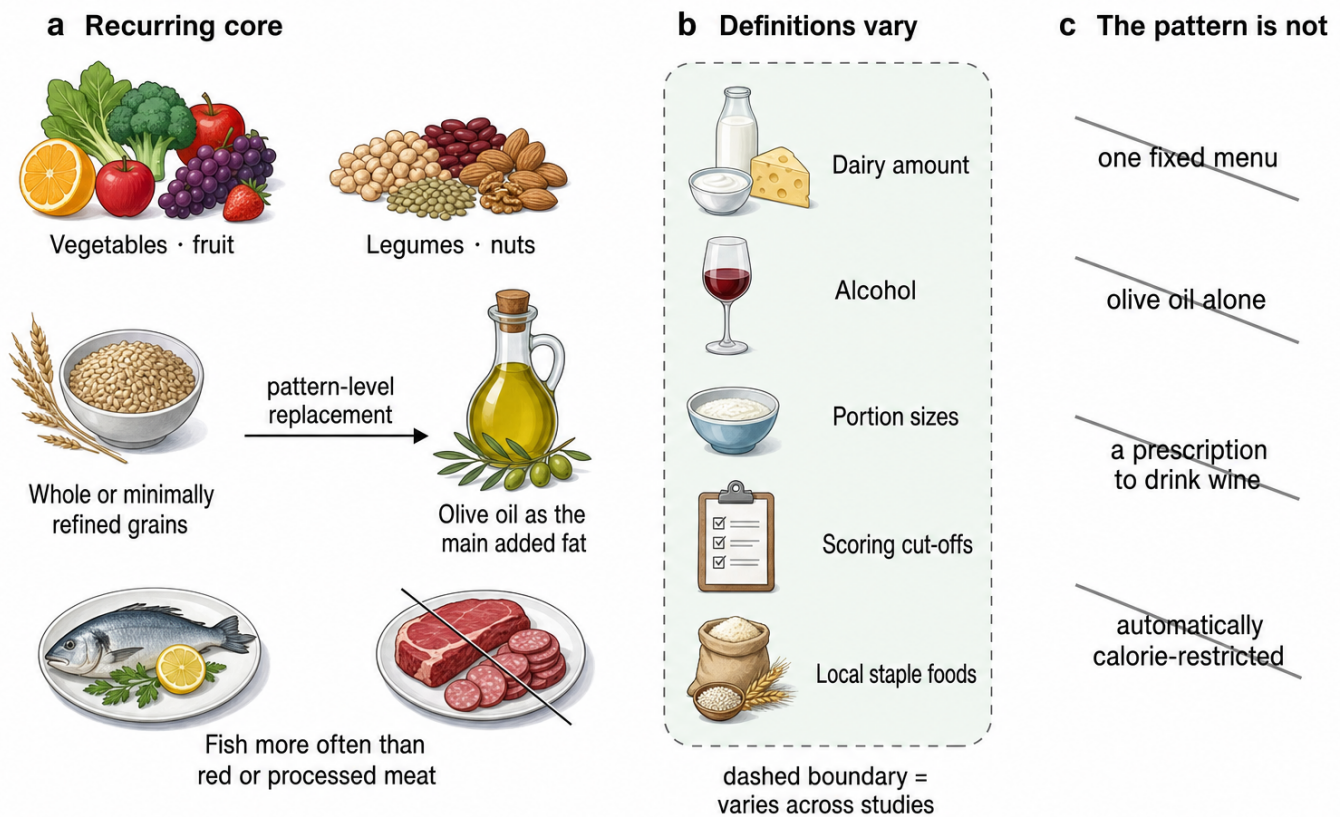


Figure 1. A Mediterranean diet is a pattern, not one fixed menu. The recurring food-group pattern is separated from definition-dependent choices and unsupported single-ingredient or fixed-menu interpretations. The replacement arrow is pattern-level synthesis, not a serving prescription or evidence that one component causes the outcomes. [Davis et al. 2015](#), [Bach-Faig et al. 2011](#), [Hutchins-Wiese et al. 2022](#), [Keshani et al. 2024](#)

This instability sets the outcome standard: effects belong to the tested pattern and support package, not to the label or one food.

03 Cardiovascular endpoint trials

PREDIMED provides the largest randomized primary-prevention dataset, but only the corrected republication is usable evidence. A statistical investigation of 5,087 randomized trials detected improbable baseline distributions, prompting scrutiny of prominent reports. [Carlisle 2017](#) PREDIMED investigators then identified household members enrolled without individual randomization and two sites with protocol departures. The 2013 report was retracted; the 2018 republication avoided assuming fully individual randomization and added exclusions and cluster-robust sensitivity analyses. [Estruch et al. 2018](#)

In the corrected analysis, 7,447 adults at high cardiovascular risk were followed for a median of 4.8 years. Compared with low-fat advice, the primary composite of myocardial infarction, stroke, or cardiovascular death had a hazard ratio (HR) of 0.69 (95% confidence interval (CI) 0.53–0.91) for the extra-virgin-olive-oil arm and 0.72 (0.54–0.95) for

the nut arm. Excluding 1,588 participants with known or suspected allocation departures yielded essentially the same estimates. [Estruch et al. 2018](#) The correction preserves a clinically important result while reducing the certainty that would attach to an entirely conventional individual randomized controlled trial.

Two secondary-prevention trials extend the finding beyond one study. The Lyon Diet Heart Study randomized 605 people after myocardial infarction and reported cardiac death or nonfatal infarction in 14 Mediterranean-pattern participants and 44 controls over 46 months. [de Lorgeril et al. 1999](#) Its effect was strikingly large and has not been reproduced at the same magnitude. CORDIOPREV randomized 1,002 patients with established coronary disease to Mediterranean or low-fat diets; after seven years, 87 versus 111 participants experienced a major event, corresponding to adjusted hazard ratios around 0.72–0.75. [Delgado-Lista et al. 2022](#) The three settings therefore agree on direction, while trial size, intervention composition, and effect magnitude differ.



TRIAL	SETTING	COMPARISON	MAIN ENDPOINT RESULT
PREDIMED republication	High-risk primary prevention; n=7,447	Olive oil or nuts vs low-fat advice	HR 0.69 and 0.72 for major cardiovascular events Estruch et al. 2018
Lyon Diet Heart	Post-infarction; n=605	Mediterranean pattern vs prudent Western diet	Cardiac death/nonfatal infarction: 14 vs 44 de Lorgeril et al. 1999
CORDIOPREV	Coronary disease; n=1,002	Mediterranean vs low-fat diet	87 vs 111 major events; adjusted HR about 0.72–0.75 Delgado-Lista et al. 2022

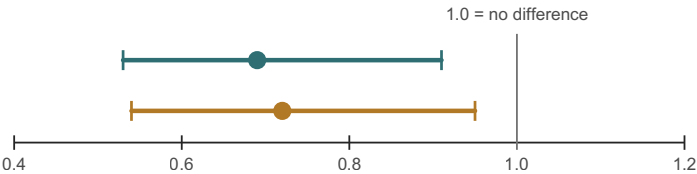
The exact trial values and the limits on comparing them are shown in [Figure 2](#).

a PREDIMED republication · 2018

High cardiovascular risk · n=7,447 · median 4.8 years
Myocardial infarction, stroke or cardiovascular death
Hazard ratio versus low-fat advice

Extra-virgin olive oil
0.69 (95% CI 0.53–0.91)

Nuts
0.72 (95% CI 0.54–0.95)



b Lyon Diet Heart

After myocardial infarction · n=605 · 46 months
Cardiac death or nonfatal myocardial infarction



c CORDIOPREV

Established coronary disease · n=1,002 · 7 years
Major cardiovascular events

Adjusted hazard ratio 0.72–0.75



Endpoints and follow-up differ · do not compare magnitudes across panels
Original 2013 PREDIMED report: retracted · corrected 2018 republication shown

95% CI = 95% confidence interval

Figure 2. Three endpoint trials support cardiovascular benefit. The proportional panel shows corrected PREDIMED hazard ratios and confidence intervals, while equal-size markers accompany the reported Lyon Diet Heart and CORDIOPREV event counts. Populations, comparators, endpoints, and follow-up differ, so cross-panel magnitude comparisons are invalid; the original 2013 PREDIMED report is retracted, and only the corrected 2018 republication is plotted. [Estruch et al. 2018](#), [de Lorgeril et al. 1999](#), [Delgado-Lista et al. 2022](#), [Karam et al. 2023](#)

These trials are the review's causal anchor, but their differences support a programme-level direction rather than one universal effect size.

04 Comparative cardiovascular evidence

The most informative modern synthesis compares supported programmes with both minimal advice and other active diets. A [network meta-analysis](#) of 40 trials and 35,548 higher-risk participants estimated that Mediterranean programmes reduced all-cause mortality (odds ratio 0.72), cardiovascular mortality (0.55), stroke (0.65), and nonfatal infarction (0.48) versus minimal intervention. At intermediate baseline risk, the estimates corresponded to about 17 fewer deaths, 13 fewer cardiovascular deaths, seven fewer

strokes, and 17 fewer infarctions per 1,000 people over five years. [Karam et al. 2023](#) However, low-fat programmes also reduced mortality and infarction, and there was no convincing advantage of Mediterranean over low-fat programmes for those endpoints. The result supports structured dietary care more strongly than dietary exceptionalism.

Earlier syntheses were more cautious. Six randomized trials with 10,950 participants produced a risk ratio of 0.63 (95% CI 0.53–0.75) for major vascular events but a null 1.00 (0.86–1.15) for all-cause mortality; excluding one trial with serious integrity concerns erased several favourable secondary findings. [Liyanage et al. 2016](#) A Cochrane review of 30 trials and 12,461 participants emphasized low-to-moderate certainty, heterogeneous definitions, sparse

events, and weak adverse-event reporting. [Rees et al. 2019](#) Another review found benefits from unrestricted-fat Mediterranean patterns but noted that clinical-event evidence rested on few trials. [Bloomfield et al. 2016](#)

Newer reviews still find fewer cardiovascular events but remain structurally dependent on PREDIMED. [Sebastian et al. 2024](#) The 2026 Italian-guideline syntheses covered 87 primary-prevention studies with more than 1.4 million participants and 19 secondary-prevention studies with more than 91,000; they reported lower incident and recurrent risk but combined randomized and cohort estimates. [Volpe et al. 2026a](#), [Volpe et al. 2026b](#) Risk-factor differences from low-fat diets are smaller than event contrasts, and active-diet differences are narrower than comparisons with usual care. [Nordmann et al. 2011](#)

The cardiovascular case is therefore strongest against minimal support, less distinctive against another supported diet, and less direct once the evidence shifts to cohort longevity.

05 Mortality and longevity

Prospective cohorts show a dose-response association with survival. Each two-point score increase corresponded to 25% lower mortality in 22,043 Greek adults and 8% in 74,607 older Europeans. [Trichopoulou et al. 2003](#), [Trichopoulou et al. 2005](#) Geographic replication thus preserved the association at a smaller magnitude.

Pooling 12 cohorts and 1.57 million participants gave an all-cause mortality risk ratio of 0.91 per two score points; later syntheses remained near a 9%–10% reduction despite varying scores and follow-up. [Sofi et al. 2008](#), [Eleftheriou et al. 2018](#), [Soltani et al. 2019](#) A 2026 meta-analysis of 54 cohorts, 1.83 million participants, and 346,034 deaths found risk ratio 0.96 (95% CI 0.95–0.97) per point, with moderate certainty. [Nucci et al. 2026](#) This graded association does not show that one menu reduces mortality by a fixed percentage.

US cohorts support transfer: top healthy-pattern categories were associated with total mortality HR 0.82 and cardiovascular disease HR 0.83. [Shan et al. 2023](#), [Shan et al. 2020](#) Among 25,315 women followed for up to 25 years, top adherence corresponded to mortality HR 0.77, only partly explained by measured metabolic and inflammatory pathways. [Ahmad et al. 2024](#) In the Moli-sani diabetes cohort, mortality HR was 0.63 per two score points. [Bonaccio et al. 2016](#)

Objective measures reduce reporting concerns but remain observational. In 642 older Italians followed for 20 years, a biomarker score predicted all-cause mortality HR 0.72 and cardiovascular mortality HR 0.55 for the highest versus lowest tertile, outperforming questionnaire scoring. Agreement was low, biomarkers were measured only at baseline, and residual confounding remained possible. [Hidalgo-Libera et al. 2021](#) Mortality evidence is therefore consistent

and dose-responsive, but its causal magnitude remains uncertain without a trial powered for that endpoint.

06 Diabetes, weight, and metabolic health

Type 2 diabetes is the non-cardiovascular outcome where experimental, prospective, and biomarker evidence align most closely. Both PREDIMED interventions lowered incidence versus low-fat advice in participants free of diabetes. [Salas-Salvadó et al. 2014](#) Up to eight annual adherence measurements yielded HR 0.80 (95% CI 0.70–0.92) per two MEDAS points and 0.46 (0.25–0.83) for highest versus lowest adherence. [Martínez-González et al. 2023](#) Because attained adherence is partly behavioural, these estimates show a gradient within a trial context.

PREDIMED-Plus tested an intensive weight-loss package. Among 4,746 adults free of diabetes, a 30%-energy-restricted Mediterranean diet plus activity reduced six-year incidence by 31% relative to unrestricted Mediterranean diet: 9.5% versus 12.0%. [Ruiz-Canela et al. 2025](#) The comparison cannot isolate diet, energy deficit, exercise, and behavioural contact. In a 1,521-person imaging substudy, the package reduced total and visceral fat and attenuated relative lean-mass decline over three years, with partial weight regain after year one. [Konieczna et al. 2023](#)

A score based on five carotenoids and 24 fatty acids identified the Mediterranean arm of a feeding trial with C-statistic 0.88, then predicted diabetes HR 0.71 (95% CI 0.65–0.77) per standard deviation in 22,202 EPIC-InterAct participants, versus 0.90 for self-report. [Sobiecki et al. 2023](#) It may capture other foods and residual confounding, but weakens reporting bias as the sole explanation. A Mediterranean-lifestyle index predicted HR 0.70 for highest versus lowest quartile in 112,493 UK Biobank adults, although it included activity, rest, and social habits. [Maroto-Rodriguez et al. 2023](#)

Prospective-study reviews estimate 19%–23% lower diabetes incidence with higher adherence, an observational result consistent with but less causal than the trials. [Schwingshackl et al. 2015](#), [Esposito et al. 2015](#) Treatment effects in established diabetes are smaller; reviews find modest glycaemic improvement versus comparator diets. Across 11 trials, glycated haemoglobin (HbA1c) was 0.31 percentage points lower, fasting glucose 0.85 mmol/L lower, and body mass index 0.83 kg/m² lower. [Wu et al. 2025](#) These are adjunctive changes, not evidence that diet replaces medication.

Weight-loss comparisons also argue against uniqueness. DIRECT found two-year losses of 4.4 kg with Mediterranean, 2.9 kg with low-fat, and 4.7 kg with low-carbohydrate diets in 322 adults. [Shai et al. 2008](#) Across 121 trials, modest six-month losses diminished by 12 months; Mediterranean programmes retained some cardiovascular risk-factor improvement but no superior long-term weight loss. [Ge et al. 2020](#) A 50-study synthesis estimated changes of about –2.4 mm Hg systolic pressure, –6

mg/dL triglycerides, -4 mg/dL glucose, and under 1 cm waist circumference; across 57 trials and 36,983 participants, direction favoured Mediterranean diet for 18 of 28 metabolic measures, but heterogeneity was often high. [Kastorini et al. 2011](#), [Papadaki et al. 2020](#) A green variant reduced visceral fat by 14.1%, versus 6.0% with standard Mediterranean diet and 4.2% with healthy guidance, but added walnuts, green tea, Mankai, energy restriction, and activity in a workforce that was 88% male. [Zelicha et al. 2022](#)

Adding olive oil does not reproduce the whole pattern: a 2024 meta-analysis of enriched Mediterranean trials found no consistent cardiometabolic advantage except a very small triglyceride reduction. [Keshani et al. 2024](#) In pre-existing metabolic disease, a 69-study synthesis found modest improvements in body mass index, waist, low-density lipoprotein (LDL) cholesterol, triglycerides, fasting glucose, insulin resistance, and C-reactive protein (CRP), but inconsistent HbA1c and lean-mass results. [Muscogiuri et al. 2026](#) Diabetes repeats the cardiovascular pattern: the strongest benefits belong to the supported diet, while weight and risk-factor changes are shared with other effective diets.

07 Cancer

Cancer is where breadth of association most clearly exceeds causal depth. An umbrella review judged evidence comparatively robust for overall cancer incidence and mortality but only suggestive or weak for many individual sites. [Dinu et al. 2018](#) The largest updated synthesis included 117 studies and 3.20 million participants. High adherence was associated with cancer mortality risk ratio 0.87, colorectal cancer 0.83, breast cancer 0.94, and larger reductions for some less common sites, yet most site-specific comparisons were rated low or very low certainty. [Morze et al. 2021](#)

Repeated meta-analyses find the same broad direction but share the same observational inputs, scoring heterogeneity, and residual lifestyle confounding. [Schwingshackl et al. 2017](#) Dose-response analysis supports lower colorectal risk, and a cohort/meta-analysis of postmenopausal breast cancer found the inverse association concentrated in estrogen-receptor-negative tumours. [Zhong et al. 2020](#), [van den Brandt & Schulpen 2017](#) These subgroup patterns can be biologically interesting without being intervention effects.

Randomized evidence consists mainly of one small secondary outcome. In PREDIMED, the olive-oil group experienced fewer invasive breast cancers than low-fat controls, HR 0.32 (95% CI 0.13–0.79), but only 35 cases occurred and cancer was not the trial's primary endpoint. [Toledo et al. 2015](#) A breast-cancer umbrella review reaches a favourable overall interpretation while acknowledging the predominance of observational evidence. [González-Palacios Torres et al. 2023](#) Olive-oil intake itself is associated with lower cardiovascular disease, diabetes, and mortality, but its pooled cancer estimate is null at risk ratio 0.94 (0.86–1.03). [Martínez-González et al. 2022](#) The cancer literature therefore sup-

ports prevention hypotheses and a generally plant-rich pattern, not a claim that Mediterranean diet treatment prevents specific cancers.

08 Cognition and dementia

Prospective studies associate Mediterranean adherence with lower cognitive decline. A 2025 meta-analysis of 23 studies reported HR 0.82 for cognitive impairment, 0.89 for dementia, and 0.70 for Alzheimer disease, while documenting heterogeneity, possible publication bias, and limited non-Western representation. [Fekete et al. 2025](#) A separate updated synthesis also found inverse dementia and Alzheimer associations. [Nucci et al. 2024](#) UK Biobank provides a useful absolute-risk estimate: among 60,298 adults, high adherence corresponded to dementia HR 0.77 and absolute risk 1.18% versus 1.73% over 9.1 years, without measurable interaction with polygenic risk. [Shannon et al. 2023](#)

Earlier reviews combined favourable cohorts with small trials. One synthesis found mild cognitive impairment risk ratio 0.75 and Alzheimer disease 0.71 in cohorts; two trials yielded an episodic-memory standardized mean difference of 0.20. [Fu et al. 2022](#) Another meta-analysis reached a similar observational conclusion. [Singh et al. 2014](#) A PREDIMED-Navarra substudy reported better global cognition after 6.5 years, but it was small and inherited the allocation limitations of the parent trial. [Martínez-Lapiscina et al. 2013](#)

The largest dedicated trial sharply lowers expectations. The three-year MIND trial randomized 604 older adults with suboptimal diets and obesity history; both intervention and control were calorie-restricted. Global cognition differed by only 0.035 standard deviations (95% CI -0.022 to 0.092), and brain magnetic resonance imaging (MRI) outcomes did not differ. [Barnes et al. 2023](#) A MIND-diet meta-analysis remains favourable because its pooled estimate is dominated by cohorts, whereas a 2024 PREVENT analysis found no association with cognition or neuroimaging. [Chen et al. 2023](#), [Gregory et al. 2024](#) The most defensible synthesis is that long-term adherence may mark lower dementia risk, but a three-year supported MIND intervention did not improve cognition in relatively healthy older volunteers.

09 Depression and mental health

Prospective cohorts associate Mediterranean adherence with lower incident depression. Meta-analyses report an inverse association of roughly one-third for the highest adherence groups, although reverse causation is difficult to exclude because early depression can alter shopping, appetite, activity, and diet quality. [Lassale et al. 2019](#), [Shafiei et al. 2019](#) The observational result is therefore compatible with benefit without proving treatment efficacy.

Small randomized trials provide a causal signal but an unstable magnitude. SMILES randomized 67 adults with major depression to dietary support or befriending-style social support for 12 weeks. The diet group improved 7.1 points

more on the [Montgomery-Åsberg Depression Rating Scale](#), and remission was 32% versus 8%. [Jacka et al. 2017](#) HELFIMED enrolled 152 adults and found greater symptom improvement with group Mediterranean education plus fish-oil supplementation than with social groups, a design that cannot isolate the diet. [Parletta et al. 2019](#) AMMEND extended the result to 72 young men over 12 weeks, again in a small, selected population. [Bayes et al. 2022](#)

Pooling magnifies the uncertainty. Five trials with 1,507 participants yielded symptom standardized mean difference -0.53 , but heterogeneity was 87%, the prediction interval crossed zero, and certainty was low. [Bizzozero-Peroni et al. 2025](#) A 2025 review identified 104 mental-health studies—88 observational and only 16 interventional—and found favourable associations with depression, anxiety, stress, quality of life, and well-being. [Kabthymmer et al. 2026](#) Direction is consistent enough to justify larger trials, but current evidence does not identify a stable antidepressant effect or support replacing established care.

10 Fatty liver

Mediterranean interventions reduce hepatic fat, but direct comparison weakens claims of unique superiority. In a 12-week randomized trial, ad libitum Mediterranean and low-fat diets both reduced steatosis without a decisive between-diet difference. [Properzi et al. 2018](#) In DIRECT, decreased hepatic fat statistically mediated part of the Mediterranean diet's cardiometabolic advantage over low fat, but mediation cannot identify the responsible foods. [Gepner et al. 2019](#)

The largest imaging effects come from an enriched, energy-restricted variant. DIRECT-PLUS randomized 294 adults to healthy guidance, standard Mediterranean, or green Mediterranean diets, with physical activity in every arm. Relative liver-fat reduction was 12.2%, 19.6%, and 38.9%, respectively, after 18 months; the green arm added walnuts, green tea, Mankai, and further red-meat restriction. [Yaskolka Meir et al. 2021](#) This shows that a complex enriched programme can change liver fat, not that olive oil alone treats liver disease.

Meta-analyses explain the mixed interpretation. Earlier randomized-trial pooling found improvement in steatosis and insulin resistance and small, inconsistent enzyme changes. [Kawaguchi et al. 2021](#), [Sangouni et al. 2022](#) A 2024 head-to-head review concluded that Mediterranean and low-fat diets had similar short-term effects on liver enzymes and fat content. [Xiong et al. 2024](#) An Asian-adapted randomized trial suggests the approach can be transferred culturally, but adaptation changes the tested foods and long-term clinical liver outcomes remain unavailable. [Chooi et al. 2024](#)

Fatty liver repeats the weight pattern: intermediate outcomes improve, but active-diet comparisons and bundled interventions limit claims of Mediterranean-specific efficacy.

11 Biological pathways

Mechanistic evidence supports plausibility through several pathways rather than one “active ingredient.” Trial syntheses report small changes in endothelial function and selected inflammatory markers, with high heterogeneity across assays and populations. [Schwingshackl & Hoffmann 2014](#), [Koelman et al. 2022](#) A 2026 review of 33 randomized trials and 3,476 participants found reductions in high-sensitivity CRP, interleukin-6, and interleukin-17, but not conventional CRP, interleukin-10, tumour-necrosis factor, or total antioxidant capacity. [Keshani et al. 2026](#) Reviews therefore implicate fat quality, plant polyphenols, fibre fermentation, inflammation, endothelial function, and insulin sensitivity without establishing one required component. [Itsiopoulos et al. 2022](#)

Controlled feeding and isocaloric designs separate dietary composition from weight loss. In an eight-week trial of 82 overweight or obese adults, a weight-maintaining Mediterranean diet lowered total cholesterol and altered faecal bile acids, microbial functions, and metabolites. Glucose, insulin, and CRP did not change. [Meslier et al. 2020](#) In older adults, the one-year NU-AGE intervention shifted microbial taxa and functions that covaried with lower frailty and inflammation; the analysed microbiome subset included 612 participants across five countries. [Ghosh et al. 2020](#) A separate obesity trial found that both Mediterranean and low-fat healthy diets modified gut communities alongside improved insulin sensitivity. [Haro et al. 2016](#)

Molecular profiles are also becoming exposure measures. Plasma metabolomic patterns tracked Mediterranean adherence and subsequent cardiovascular risk, though they cannot by themselves prove mediation. [Li et al. 2020](#) In DIRECT-PLUS, baseline bile acids and particular microbial species modified lipid and adiposity responses, while a PREDIMED-Plus analysis connected intervention-related microbiome and metabolome shifts with risk-factor changes. [Gao et al. 2024](#), [García-Gavilán et al. 2024](#) These studies clarify how people may respond differently; they do not upgrade surrogate changes into evidence for fewer cancers, dementia cases, or deaths.

12 Implementation, cost, and sustainability

Transfer requires adapting the plant-forward structure and fat quality to local staples, availability, cooking, and culture. [Hoffman & Gerber 2013](#), [Woodside et al. 2022](#) Scores built around olive oil, wine, and regional medians may misclassify an equally plant-rich pattern elsewhere. Implementation concerns what replaces refined grains, processed meat, and saturated fat, not geographic authenticity.

The intervention package, adherence conditions, and the causal stopping point are traced in [Figure 3](#).

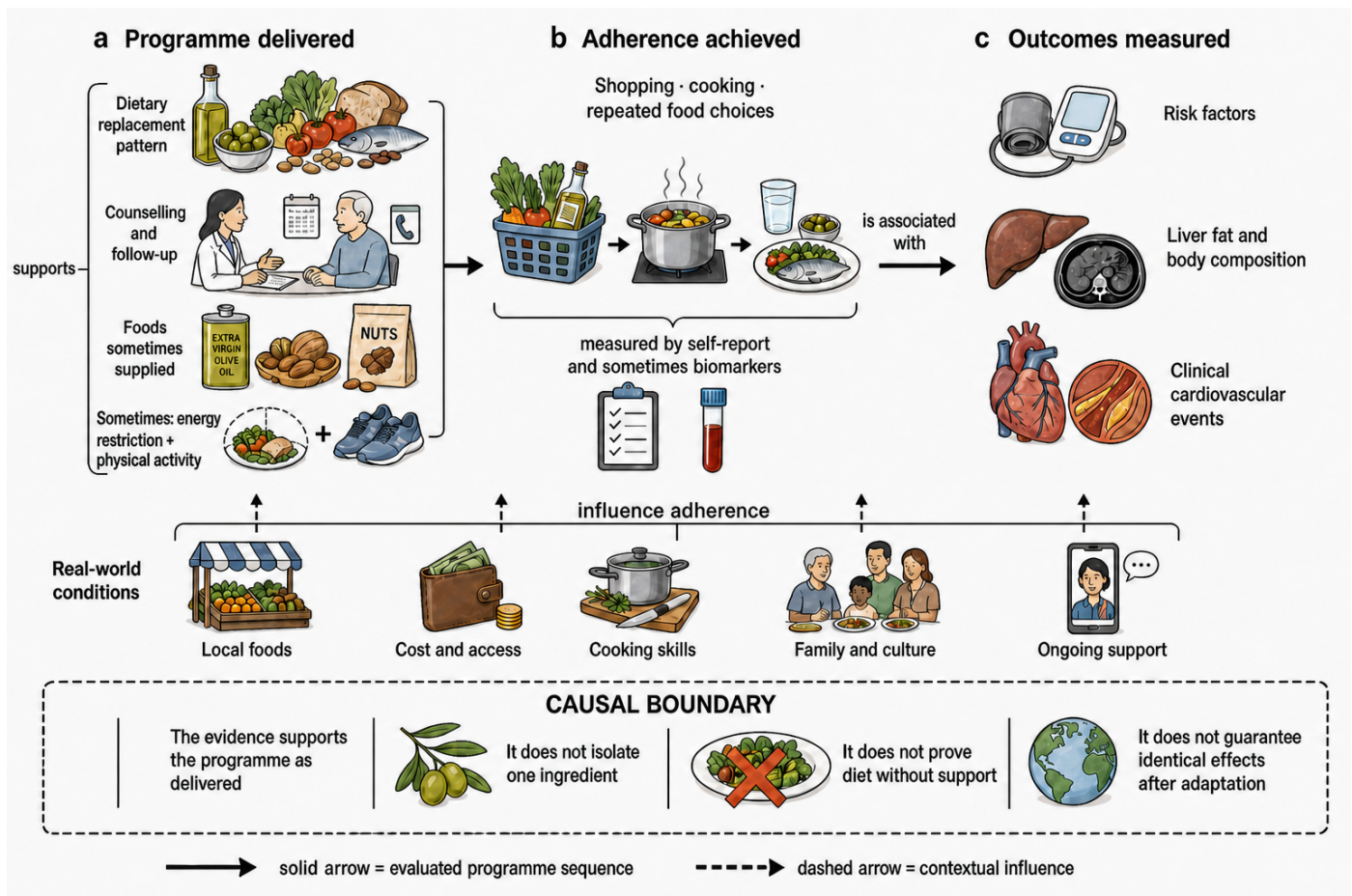


Figure 3. Benefits belong to the supported programme as delivered. Trials evaluate a replacement pattern with varying counselling, food provision, and lifestyle support; real-world conditions influence adherence before risk factors, imaging outcomes, or clinical events are measured. The arrows express programme logic rather than a quantified mediation model, and the dashed boundary prevents attributing the package to one ingredient, unsupported diet alone, or identical effects after cultural adaptation. [Estruch et al. 2018](#), [Jennings et al. 2024](#), [Biggi et al. 2024](#), [Lorca-Camara et al. 2024](#), [Ruiz-Canela et al. 2025](#)

Implementation is part of efficacy: trial benefits belong to supported replacement patterns and may change with support, affordability, or cultural fit.

Supported efficacy exceeds unsupported adoption. MedEx-UK improved adherence and acceptability in older adults at dementia risk but included behavioural maintenance and physical activity. [Jennings et al. 2024](#) Cost, access, cooking skills, family norms, and knowledge remain barriers. [Biggi et al. 2024](#) PREDIMED supplied oil or nuts; PREDIMED-Plus added dietitian contact and activity coaching. [Estruch et al. 2018](#), [Ruiz-Canela et al. 2025](#) Benefits may decline when those supports disappear.

Economic and environmental effects depend on assumptions. A secondary-prevention model found Mediterranean diet plus activity cost-effective but inherited adherence and

event estimates from other data. [Bonekamp et al. 2024](#) Of 35 sustainability studies, mostly models, 91% favoured the pattern environmentally; gains centred on less ruminant meat, while water demand and local production changed rankings. [Lorca-Camara et al. 2024](#) Med Diet 4.0 likewise treats health, environment, culture, and local economy as linked rather than interchangeable. [Dernini et al. 2017](#)

Alcohol requires a modern qualification. Traditional scores often award a point for moderate wine, but alcohol raises cancer, dependence, and injury risk. A contemporary review argues that wine is not necessary to preserve the pattern's benefits and should not be recommended to non-drinkers. [Martínez-González 2024](#) The therapeutic interpretation is therefore vegetables, legumes, whole grains, nuts, fish, and unsaturated fats—not an instruction to begin drinking.



13 Overall certainty

OUTCOME	CURRENT CONCLUSION	MAIN REASON CONFIDENCE IS LIMITED
Major cardiovascular events	Probably reduced in higher-risk adults, particularly versus minimal support	Few endpoint trials; corrected PREDIMED carries much of primary-prevention weight Karam et al. 2023
Type 2 diabetes	Probably prevented; established disease improves modestly	Intensive programmes combine diet, energy restriction, activity, and support Ruiz-Canela et al. 2025
All-cause mortality	Consistent 4% lower risk per adherence point in cohorts	No randomized mortality confirmation; residual healthy-user bias Nucci et al. 2026
Weight and risk factors	Small favourable changes	Similar benefits occur with other supported healthy diets Ge et al. 2020
Cancer	Several inverse site-specific associations	Predominantly observational; one small randomized secondary signal Morze et al. 2021
Cognition	Cohorts favourable, largest dedicated trial null	Active control, limited duration, exposure confounding Barnes et al. 2023
Depression	Small trials favourable	Low certainty, high heterogeneity, weak blinding Bizzozero-Peroni et al. 2025
Fatty liver	Liver fat decreases	Short-term equivalence to low-fat diets; few clinical outcomes Xiong et al. 2024

Directness declines from randomized cardiovascular and diabetes evidence through cohort mortality, predominantly observational cancer and dementia evidence, limited depression trials, and surrogate liver outcomes. Mechanisms

make the cardiometabolic findings coherent but cannot close those causal gaps.

This uneven directness and certainty are mapped in [Figure 4](#).

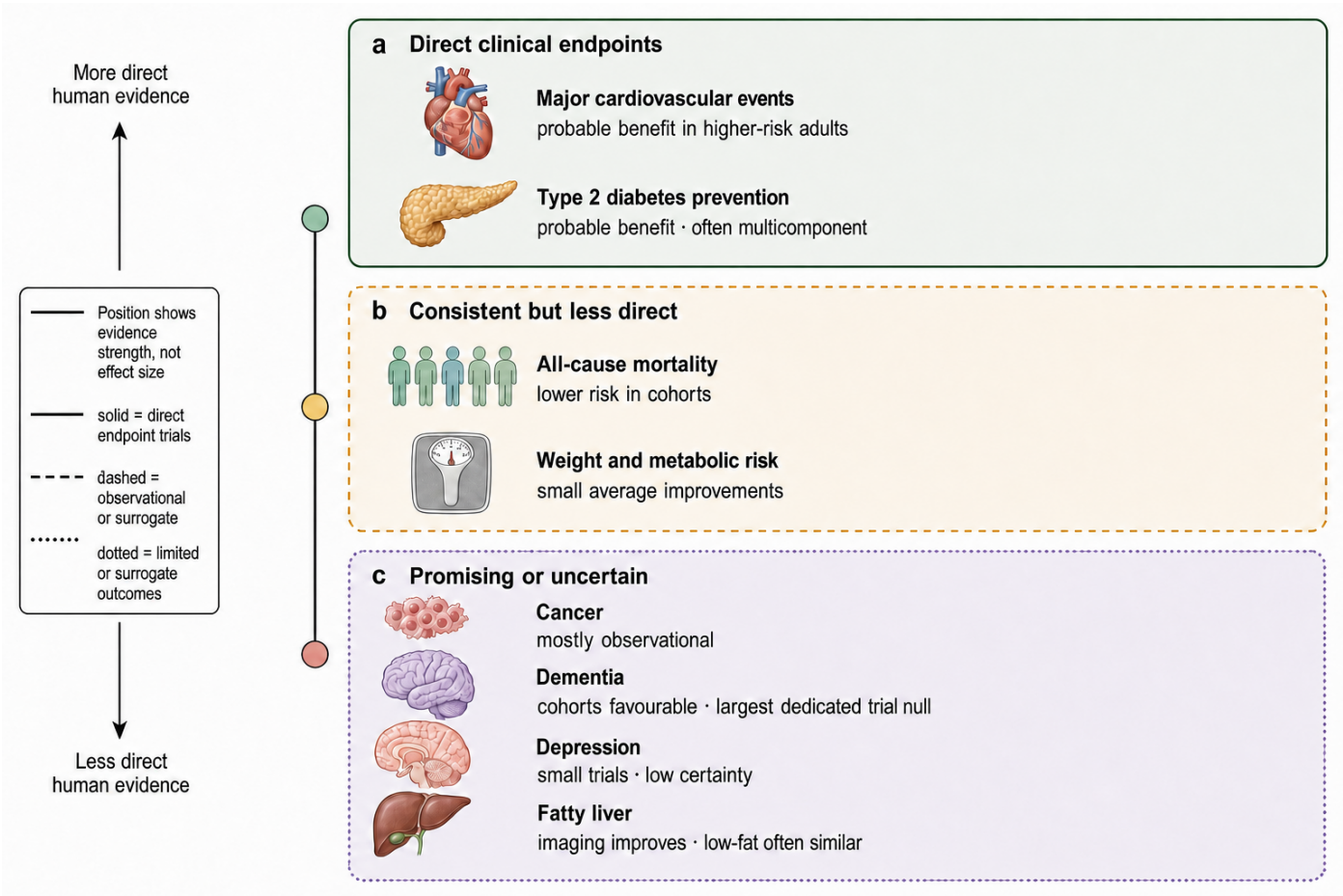


Figure 4. The evidence is strongest for cardiometabolic outcomes. Outcomes are arranged by human evidence directness and certainty, from clinical endpoint trials through consistent but less-direct evidence to limited or surrogate outcomes. This is a narrative-review synthesis rather than a formal GRADE score: position, marker area, and colour do not encode effect size or an individual's expected benefit. [Karam et al. 2023](#), [Ruiz-Canela et al. 2025](#), [Nucci et al. 2026](#), [Morze et al. 2021](#), [Barnes et al. 2023](#)

Across outcomes, greater distance from randomized clinical endpoints requires greater qualification: plausibility extends beyond demonstrated clinical benefit.

14 Conclusion

The cross-cutting pattern is an evidence gradient, not one verdict on a diet. Supported Mediterranean programmes probably reduce major cardiovascular events in higher-risk adults and prevent type 2 diabetes, especially compared with minimal advice. Mortality, cancer, and dementia conclusions rely increasingly on cohorts, while mechanisms and imaging support plausibility without proving disease prevention. Supported low-fat diets also narrow head-to-head differences, and several favourable interventions bundle

diet with energy restriction, activity, food provision, or counselling.

The defensible recommendation is therefore a culturally workable, plant-forward replacement pattern with credible cardiometabolic value, not isolated foods or nutrients treated as medicines. Depression remains a heterogeneous small-trial signal, and liver-fat improvement has not established Mediterranean-specific clinical benefit. Long-term randomized trials with active comparators, clinical endpoints, and intervention arms that separate diet composition from energy restriction, activity, and behavioural contact would show how much of the wider promise is causal and how much belongs to adherence support.

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